

Vamsi Manda

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EDUCATION

University of Florida, Florida, United States

August 2023 - May 2025

Master's, Computer Science

GPA 3.90/4

Veer Surendra Sai University of Technology, Odisha, India

August 2019 - May 2023

Bachelor of Technology, Electrical Engineering

CGPA 8.28/10

SKILLS

Programming Languages: Python, Java, C, C++, GoLang, Swift, HTML/ CSS, JavaScript, MongoDB, PHP, jQuery, XML, MYSQL, NoSQL.

Skills: Data Science, Machine Learning, AI, LLM, GENAI, Data Mining, ETL, SQL, Django, PowerBI, Tableau, CI/CD, React.js, Node.js.

Tools & Frameworks: Spacy, Spark, PyTorch, TensorFlow, OpenCV, Flask, RAG, LangChain, Streamlit, RNN, NLP, GAN, K-shot, GIT.

EXPERIENCE

Machine Learning Engineer

August 2024- Present

University of Florida

Gainesville, Florida

- Developed a distributed deep learning pipeline with **PyTorch, TensorFlow, and DDP** to train a **4D CNN** for kidney and liver motion tracking, enhancing real-time organ displacement assessment for chemotherapy planning.
- Designed a hybrid model combining **CNNs and LSTMs**, using **Seq2Seq** modelling and attention mechanisms to improve motion prediction accuracy in dynamic medical imaging.
- Optimized training performance on multi-GPU clusters by implementing **gradient accumulation**, mixed-precision training, and **GPU memory profiling**, ensuring efficient deep learning model deployment for adaptive radiotherapy.

Machine Learning Engineer

May 2021 – December 2022

Coratia Technologies

Rourkela, India

- Engineered an underwater image processing pipeline using **Python, OpenCV, and NumPy**, integrating **LABStretching** and **RGHS** to enhance **500+** images, resulting in a **35-40%** improvement in **PSNR** and **SSIM** visibility metrics.
- Optimized the enhancement workflow with **RGB Equalization** and **histogram stretching**, automating batch processing to reduce computation time by **30%** and increasing throughput to **20 images per second**.
- Integrated advanced color correction algorithms to mitigate underwater haze and color shifts, improving contrast by **25%** and color fidelity by **15%**, validated via analysis against standard datasets.

Machine Learning Intern

August 2020 – February 2021

SWAN Labs, IIT Kharagpur

Kharagpur, India

- Engineered a **Raspberry Pi IoT** module to collect environmental data (NO2, CO2, PM2.5, PM10) for **differential AQI analysis** at a traffic intersection.
- Developed and trained a regression model (**85% accuracy**) in Python to predict AQI levels using real-time sensor data and deployed a cloud-based alert system with **MQTT** and **REST APIs**, reducing response times by **30%**.

ACADEMIC PROJECTS

DeepLens: AI-Powered Real-Time Image Captioning and Deployment System

- Designed & developed an Image Captioning System using **ResNet50, VGG16, and LSTM**, improving BLEU score by **15%**.
- Built & deployed a Flask-based AI Web App, reducing latency by **30%** and efficiently handling **10K+ requests**.
- Optimized & scaled a data pipeline for real-time captioning, boosting training efficiency by **20%** on **10K+** image datasets.

AI Legal Counsel: AI Agent Legal Document Analysis System

- Developed an AI-driven legal document processing system** leveraging multi-agent collaboration with Gemini and xAI, achieving **85% accuracy** in contract analysis, risk assessment, and legal research across **10,000+ case documents**.
- Optimized legal case retrieval** by developing a **ChromaDB + FAISS + BERT** system, improving citation lookup speed by **40%** via efficient document chunking.
- Designed a **Streamlit UI** with **FastAPI** for real-time legal document uploads, cutting manual review time by **60%**.

JobGenieAI: Career Coaching Assistant

- Built an AI-driven resume analyser using **DeepSeek** API to process resumes and provide personalized feedback.
- Developed an interactive mock interview system that generated AI-powered, role-specific questions, improving user confidence and increasing interview **success rates by 35%**.

Achievements and Certification

- Won 1st prize in International Space Week by NASA and ISRO for an automated bomb detection bot prototype.
- AWS Certified Solutions Architect Associate and OCI Certified Gen AI Specialist.